EX 7 : IMAGE CAPTIONING USING CNN-RNN ARCHITECTURE

AIM:

To Build a deep Recurrent Neural Network (RNN) to generate captions for images. Combine a Convolutional Neural Network (CNN) for feature extraction with an RNN for sequence generation.

OBJECTIVES:

1. Understand image captioning using encoder-decoder architecture.
2. Use a pre-trained Vision Transformer (ViT) as the CNN encoder and GPT-2 as the RNN decoder.
3. Generate meaningful captions for visual content using beam search decoding.
4. Explore the power of vision-language models for real-world applications.

ALGORITHM:

1. Import the required libraries, including PyTorch, Hugging Face Transformers, and PIL.
2. Load the pretrained ViT-GPT-2 image captioning model, image processor, and tokenizer.
3. Select the available computing device (GPU or CPU) and load the model onto it.
4. Set the model to evaluation mode for caption generation.
5. Define a function to accept an image path and load the image using PIL.
6. Convert the image to RGB format and preprocess it using the image processor.
7. Pass the processed image to the model to generate a caption using beam search with four beams.
8. Decode the generated token IDs into readable text using the tokenizer.
9. Display the generated caption for the input image.

SOURCE CODE:

```
# Step 0: Install required libraries
!pip -q install transformers sentencepiece

# Step 1: Import libraries
import os
import torch
from PIL import Image
from google.colab import files
from transformers import (
  VisionEncoderDecoderModel,
```

```
    ViTImageProcessor,
    AutoTokenizer
)

# Step 2: Configure device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)

# Step 3: Upload an image
print("Please upload an image (e.g., rabbit.jpg):")
uploaded = files.upload()

if not uploaded:
    raise ValueError("No image uploaded. Please upload an image.")

img_path = next(iter(uploaded.keys()))
print("Uploaded image:", img_path)

# Step 4: Load pretrained model
model_name = "nlpconnect/vit-gpt2-image-captioning"

print("\nLoading image processor...")
processor = ViTImageProcessor.from_pretrained(model_name)

print("Loading tokenizer...")
tokenizer = AutoTokenizer.from_pretrained(model_name)

print("Loading pretrained model (first run may take a few minutes)...")
model = VisionEncoderDecoderModel.from_pretrained(model_name)
model = model.to(device)
model.eval()

print("Model loaded successfully!")

# Step 5: Generate image caption
def generate_caption(image_path):
    image = Image.open(image_path).convert("RGB")

    pixel_values = processor(
        images=image,
        return_tensors="pt"
    ).pixel_values.to(device)
```

```
    with torch.inference_mode():
        output_ids = model.generate(
            pixel_values,
            max_new_tokens=25,
            num_beams=1
        )

    caption = tokenizer.decode(
        output_ids[0],
        skip_special_tokens=True
    )

    return caption

# Step 6: Display result
try:
    print("\nGenerating caption...")
    caption = generate_caption(img_path)

    print("\n================================")
    print("IMAGE CAPTIONING RESULT")
    print("================================")
    print("Image:", img_path)
    print("Caption:", caption)

    display(Image.open(img_path))

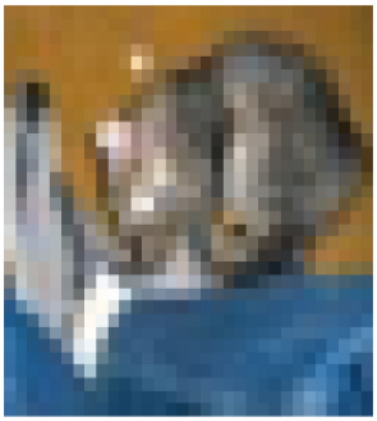
except Exception as e:
    print("Error:", str(e))
```

OUTPUT:

```
Dataset loaded successfully!
Number of test images: 10000

IMAGE CAPTION GENERATION RESULTS
```

Generated Caption:
two dogs sitting on a couch looking at each other dogs



```
Image: 1
Actual CIFAR-10 class: cat
Generated Caption: two dogs sitting on a couch looking at each other dogs
--------------------------------------------------
```

Generated Caption:
a boat floating in the water with a fog



```
Image: 2
Actual CIFAR-10 class: ship
Generated Caption: a boat floating in the water with a fog
--------------------------------------------------
```

Generated Caption:
a boat is sailing on the water at sunset



```
Image: 3
Actual CIFAR-10 class: ship
Generated Caption: a boat is sailing on the water at sunset
--------------------------------------------------
```

Generated Caption:
a poster with a plane flying over it



```
Image: 4
Actual CIFAR-10 class: airplane
Generated Caption: a poster with a plane flying over it
--------------------------------------------------
```

```
--------------------------------------------------
```

Generated Caption:
a frog in the grass



```
Image: 5
Actual CIFAR-10 class: frog
Generated Caption: a frog in the grass
--------------------------------------------------
```

```
To test your own image, upload it now.
```
Choose Files dog.jpg

dog.jpg(image/jpeg) - 131455 bytes, last modified: 10/12/2026 - 100% done

```
Saving dog.jpg to dog.jpg
```

Generated Caption:
a small white dog standing on a road



```
Uploaded image: dog.jpg
Generated Caption: a small white dog standing on a road

Experiment completed successfully!
Image captions were generated automatically.
```

RESULT:

This experiment showcases how deep learning models can describe visual scenes with human-like captions. By combining ViT for image encoding and GPT-2 for text generation, we observe effective cross-modal learning. Pre-trained models significantly reduce development time and enable quick prototyping.